

Draft v0.4 - Chapter 4: Body Composition Analysis

Personal Weight Regulation Model / N-of-1 Longitudinal Case Study

4.1 Dataset

Item	Value
Observation period	2025-09-28 to 2026-06-25
Duration	270 days
Source	Health_Analytics_Database_DailySummary.xlsx / Daily Summary
Primary variables	Weight kg; Fat mass kg; Lean mass kg
Smoothing used for figures	7-day rolling mean

4.2 Initial vs Final Body Composition

Metric	Start	End	Delta kg	Delta %	kg/month	g/week
Weight	65.58	69.63	4.05	6.2	0.457	105.0
Fat Mass	17.55	19.73	2.18	12.4	0.246	56.5
Lean Mass	48.03	49.9	1.87	3.9	0.211	48.5

Result: total weight increased while both Fat Mass and Lean Mass increased. Fat Mass accounted for the larger share of total weight change.

4.3 Monthly Body Composition

Date	Weight	Fat Mass	Lean Mass
2025-09	65.58	17.55	48.03
2025-10	66.47	18.01	48.46
2025-11	66.39	17.76	48.63
2025-12	66.59	17.75	48.84
2026-01	66.9	17.9	49.0
2026-02	67.39	18.2	49.18
2026-03	67.32	18.27	49.05
2026-04	67.91	18.69	49.22
2026-05	67.12	18.2	48.92
2026-06	68.29	19.18	49.11

4.4 Trajectories

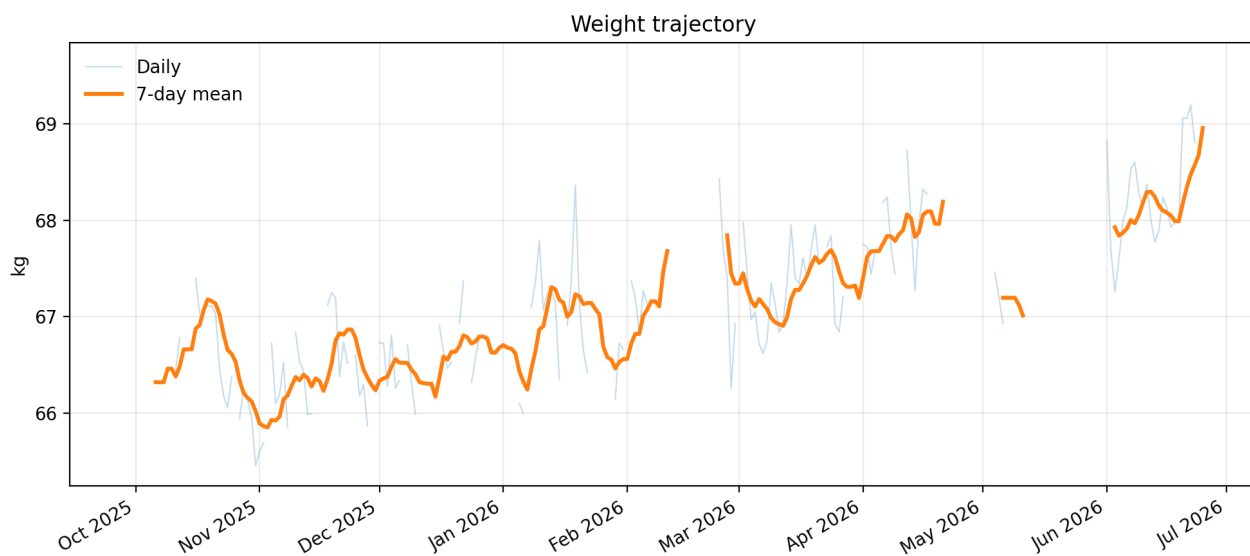


Figure 4.1. Weight trajectory, daily values and 7-day rolling mean.

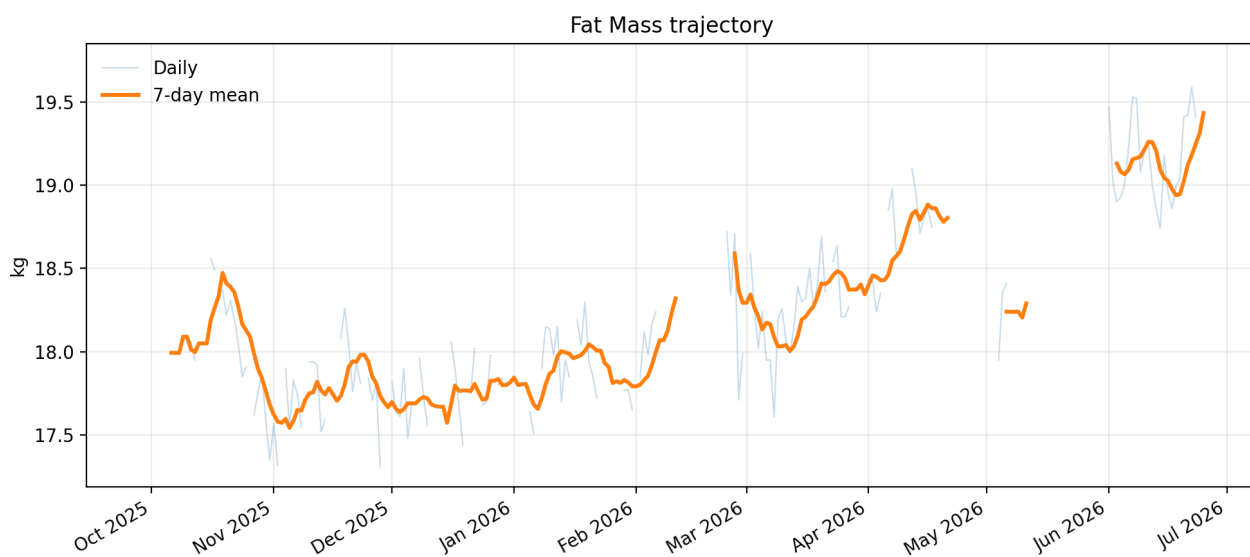


Figure 4.2. Fat Mass trajectory, daily values and 7-day rolling mean.

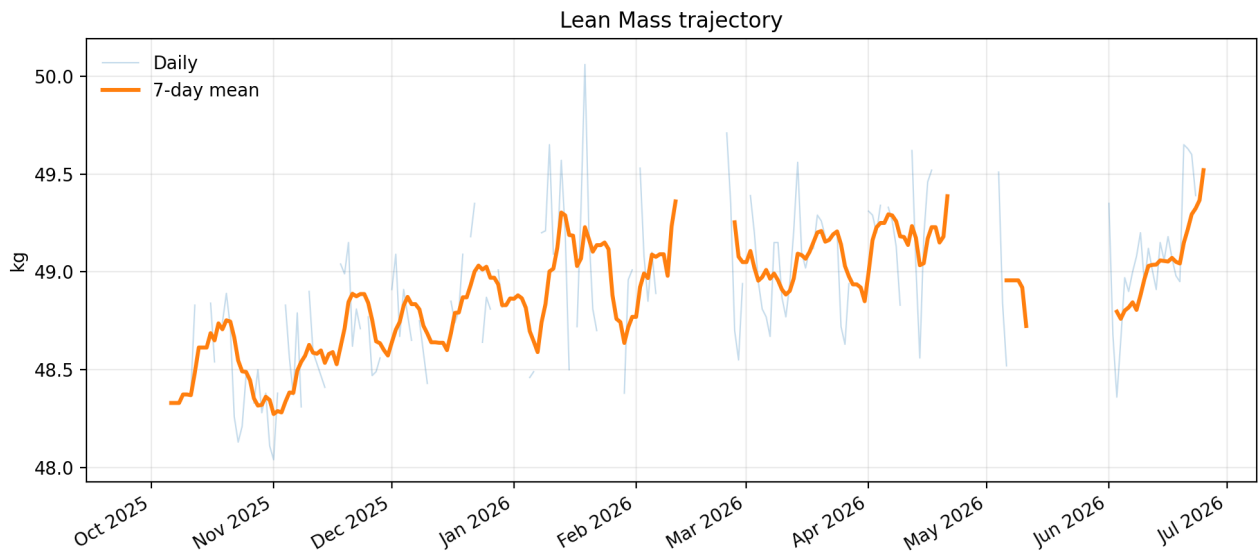


Figure 4.3. Lean Mass trajectory, daily values and 7-day rolling mean.

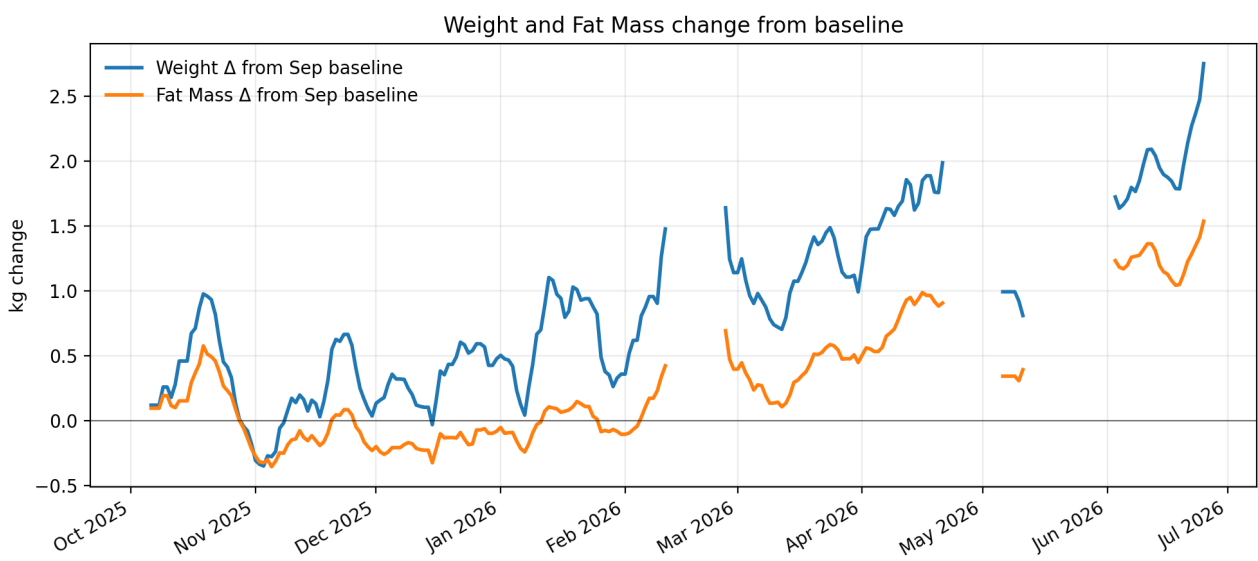


Figure 4.4. Weight and Fat Mass change relative to September baseline.

4.5 Phase Analysis

Phase	Start	End	Days	Weight Δ kg	Fat Δ kg	Lean Δ kg
Phase I: early drift / lean-dominant adaptation	2025-09-28	2026-01-31	126	1.08	0.1	0.98
Phase II: mixed transition	2026-02-02	2026-04-17	89	0.9	0.91	-0.01
Phase III: fat-dominant period	2026-05-04	2026-06-25	56	2.17	1.78	0.39

Interpretation: the body composition trajectory is not uniform across the study period. Early change was more lean-mass associated; later change was more fat-mass associated.

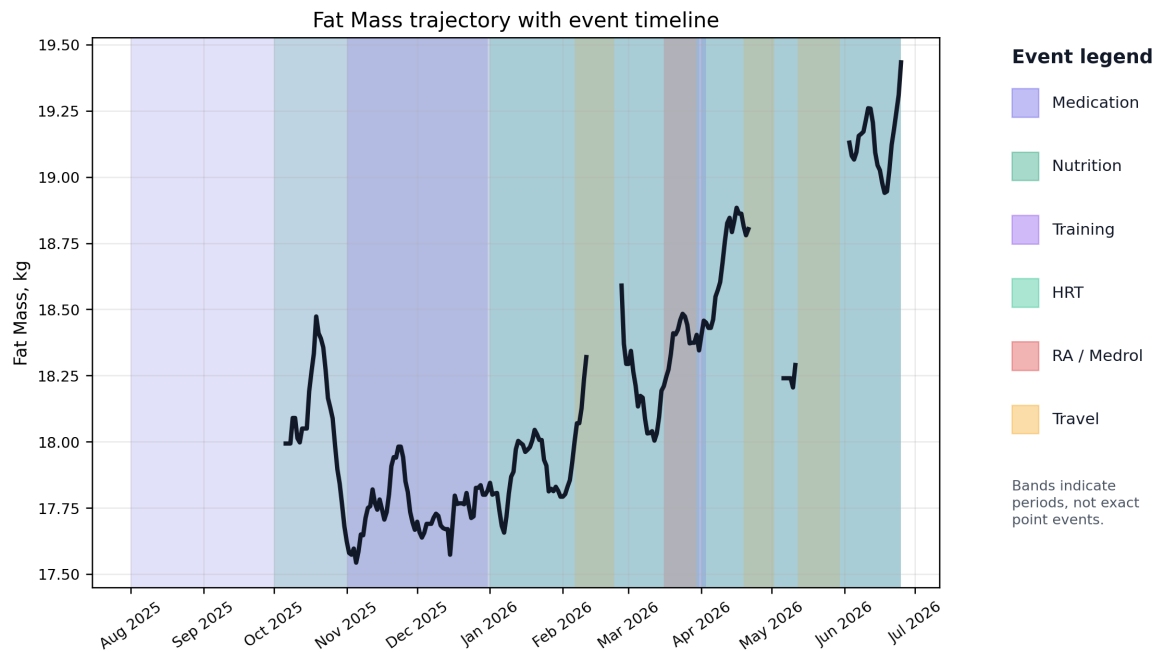


Figure 4.5. Fat Mass with major events and intervention periods. Colored vertical bands indicate event/intervention periods; the right-side legend table maps colors to categories.

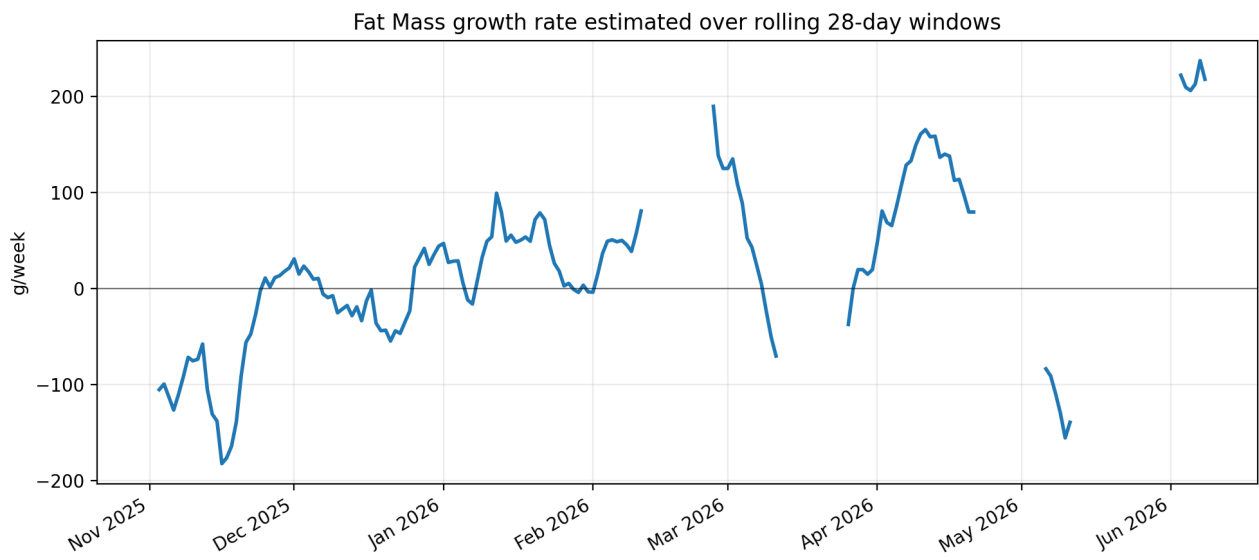


Figure 4.6. Rolling 28-day Fat Mass growth rate, expressed as g/week.

4.6 Evidence Summary

Finding	Evidence	Confidence
Fat Mass increased	Withings trend, daily data	High for direction; moderate for exact kg
Lean Mass increased	Withings trend, daily data	High for direction; moderate for exact kg
Weight change is mixed composition	Fat and Lean both increased	High
Later period became more fat-dominant	Phase analysis	Moderate